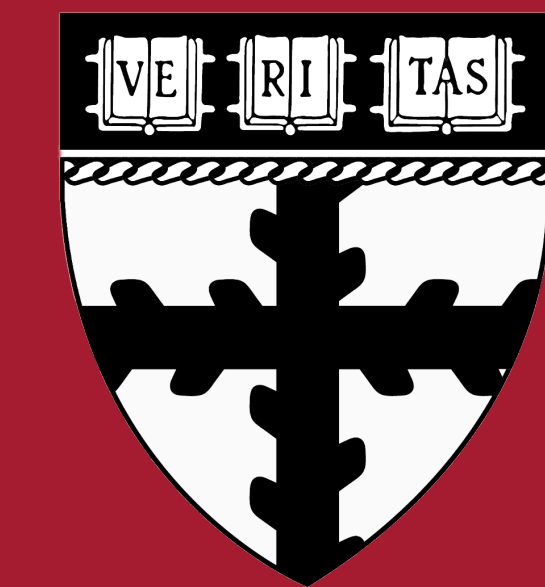




Variational Boosting for Physics-Informed Neural Networks

Kaylee Vo Pavlos Protopapas

Harvard John A. Paulson School of Engineering and Applied Sciences



Introduction

Physics-Informed Neural Networks (PINNs) solve differential equations by minimizing a residual functional $\mathcal{L}(u) = \|F(u)\|^2$ over a neural parameterization u_θ . Monolithic PINNs, however, suffer from **ill-conditioning**, **spectral bias**, and **optimization instability** when a single network must learn all scales of a solution at once.

We introduce a variational boosting framework [2] in which solutions are constructed additively in function space. Each stage trains a weak learner whose converged correction satisfies a local orthogonality condition, equivalent to a **projected functional gradient descent step** onto the tangent space of the network's function manifold. Because each correction network is deliberately small, the restricted minimization admits **full Newton** or **conjugate gradient** (CG) updates.

Motivation

The functional gradient of the residual is $\nabla_u \mathcal{L}(u) = 2 F'(u)^* F(u)$, where $F'(u)$ is the Fréchet derivative of the operator F and $F'(u)^*$ its adjoint. For nonlinear operators, this linearization evolves as u changes during training, so the curvature of the loss landscape shifts throughout optimization.

Boosting addresses this by separating the learning task across stages: each small network is individually well-conditioned, avoiding the need to learn all scales simultaneously.

Contributions

1. A theoretical, geometric interpretation of multi-stage PINNs as **projected functional descent step**.
2. **Second-order optimization** using full Newton and CG.
3. **Transfer learning** across correction stages.

Boosted PINN Framework

The solution is built as an **additive ensemble**. Stage 0 is a standard PINN trained on the full problem, giving $u^{(0)} = h_0$. Each subsequent stage $k \geq 1$ trains a small weak learner h_k to reduce the residual left by the current ensemble, and updates:

$$u^{(k)} = u^{(k-1)} + \alpha_k h_k, \quad (1)$$

where $\alpha_k > 0$ is a shrinkage parameter. After K stages, $u^{(K)} = h_0 + \sum_{k=1}^K \alpha_k h_k$. The steepest descent direction at $u^{(k-1)}$ is

$$-\nabla_{\mathcal{H}} \mathcal{L}(u^{(k-1)}) = -2 F'(u^{(k-1)})^* r^{(k-1)}, \quad (2)$$

where $r^{(k-1)}$ is the current residual and $F'(u^{(k-1)})^*$ is the adjoint of the linearized operator. Training h_k to minimize the updated residual is *approximately* equivalent to finding the element of $\mathcal{U}_k$ that best matches this steepest-descent direction.

- Linearizing F and minimizing the resulting residual gives a Gauss–Newton step, which generally differs from steepest descent.
- Training directly on the full nonlinear residual is neither, since F is re-evaluated at the moving iterate throughout training.

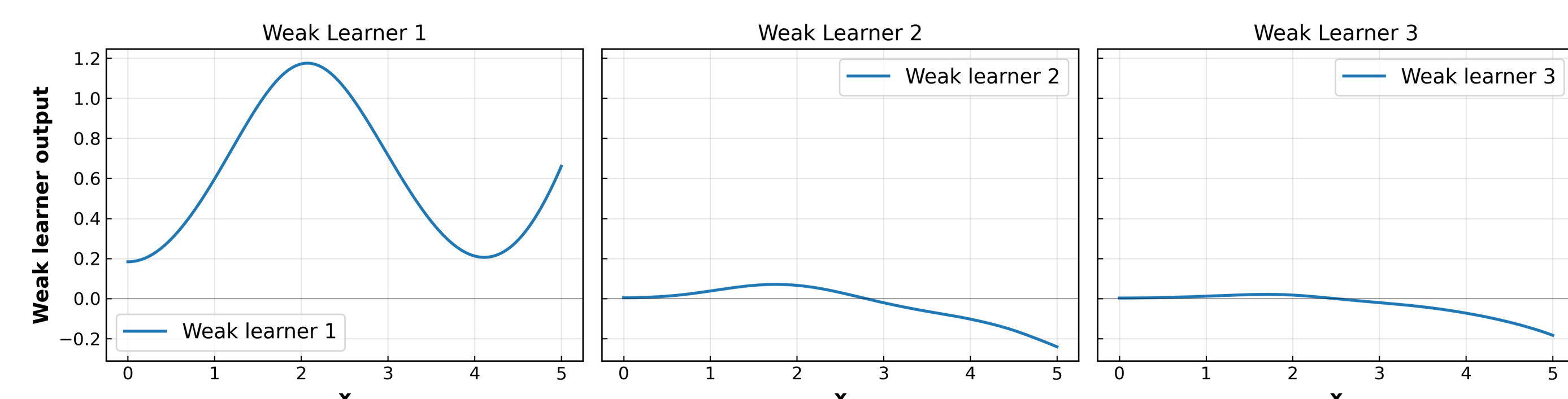


Figure 1. Weak learners h_k shrink toward zero as the residual is reduced (Duffing equation).

Second-Order Optimization

- **Conjugate Gradient**: solves $(H + \gamma I)d = -g$ using only Hessian–vector products, avoiding explicit formation of H .
- **Newton's Method**: computes H explicitly, uses Tikhonov regularization when H is not positive-definite, and applies trust-region step-size control..
- Implementation: **Adam** $\rightarrow$ **second-order optimizer** ($\sim 90\%$ Adam, then CG/Newton) lets Adam reach a well-conditioned region before exploiting local curvature.

Metric	Monolithic PINN	Boosted PINN	
	L-BFGS	Adam + CG	Adam + Newton
Residual Norm	3.88×10^{-3}	5.43×10^{-3}	3.24×10^{-3}
MSE	3.77×10^{-4}	7.08×10^{-4}	1.44×10^{-4}
Relative L^2 Error	6.19×10^{-4}	9.64×10^{-4}	2.17×10^{-4}
Training Time (s)	0.31	7.03	187.72
Number of Iterations	132	738	407
Learning Rate	1.00	0.01	0.01
Collocation Points	1,000	1,000	1,000
Network Size	2,209	2,209	2,209

Table 1. Monolithic vs. Boosted PINN performance on the Duffing equation.

Warm-Start Via Transfer Learning

Each h_k is initialized by transferring weights from h_{k-1} , with the final two layers re-initialized (Xavier) and rescaled by a stage-specific factor ξ_k . Removing transfer learning significantly degrades performance — e.g. MSE on Lotka–Volterra worsens from 1.76×10^{-3} to 7.91×10^{-1} ($p = 0.0042$).

Results Summary

Across ODE, coupled-ODE, and PDE benchmarks, the boosted PINN **matches or exceeds** standard (monolithic) PINN accuracy and converges where standard PINNs **fail outright**, an advantage especially pronounced in stiff regimes.

Problem	Std. PINN MSE	Boosted MSE	Converges?
Duffing	3.77×10^{-4}	1.44×10^{-4}	Both
NRD ($\kappa=100$, stiff)	9.27×10^{-1}	8.01×10^{-3}	Boosted only
Van der Pol ($\mu=3$)	—	5.71×10^{-3}	Boosted only
Lotka–Volterra	1.74×10^{-2}	2.92×10^{-3}	Boosted only
Burgers' ($\nu=0.002$)	9.27×10^{-1}	5.36×10^{-2}	Neither
Allen–Cahn (D=0.0001)	1.41×10^{-2}	7.10×10^{-3}	Boosted only

Table 2. Selected MSE comparison (best optimizer per model). “—” = did not converge.

Problem	Method	Iterations	Wall-Clock
Duffing	Adam	$2.18 \times$	$0.40 \times$
	Adam + CG	$2.07 \times$	$0.46 \times$
	Adam + Newton	$3.76 \times$	$0.02 \times$
Allen–Cahn (nonstiff)	Adam	$1.00 \times$	$1.26 \times$

Table 3. Convergence metrics for boosted PINN vs. standard, restricted to problems where both converged (Higher favors boosted PINN).

Standard PINN failed to converge on: NRD equation (both stiff and nonstiff), Van der Pol equation, Lotka–Volterra coupled ODE system, Allen–Cahn equation (stiff), Burgers' equation (stiff and nonstiff). Our **method converges quicker** than other ensemble methods [1], uses **smaller models**, and yields performance on the same order of magnitude.

Stiff PDE Example (Allen–Cahn)

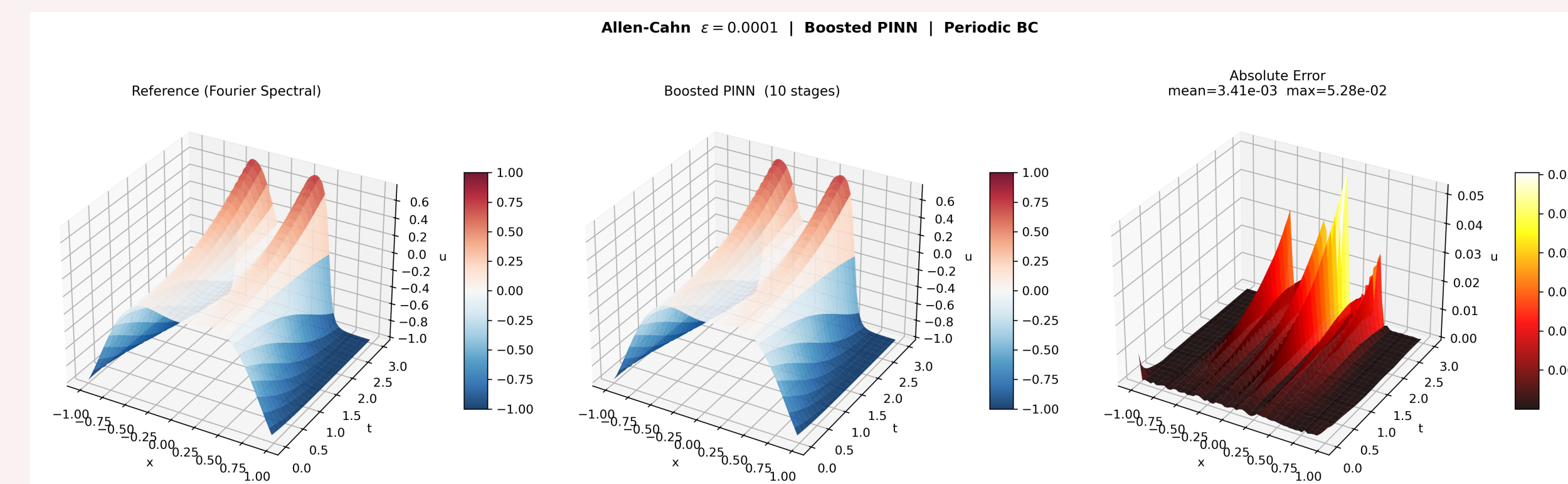


Figure 2. Boosted PINN solution cross-sections to the Allen–Cahn PDE (D=0.0001).

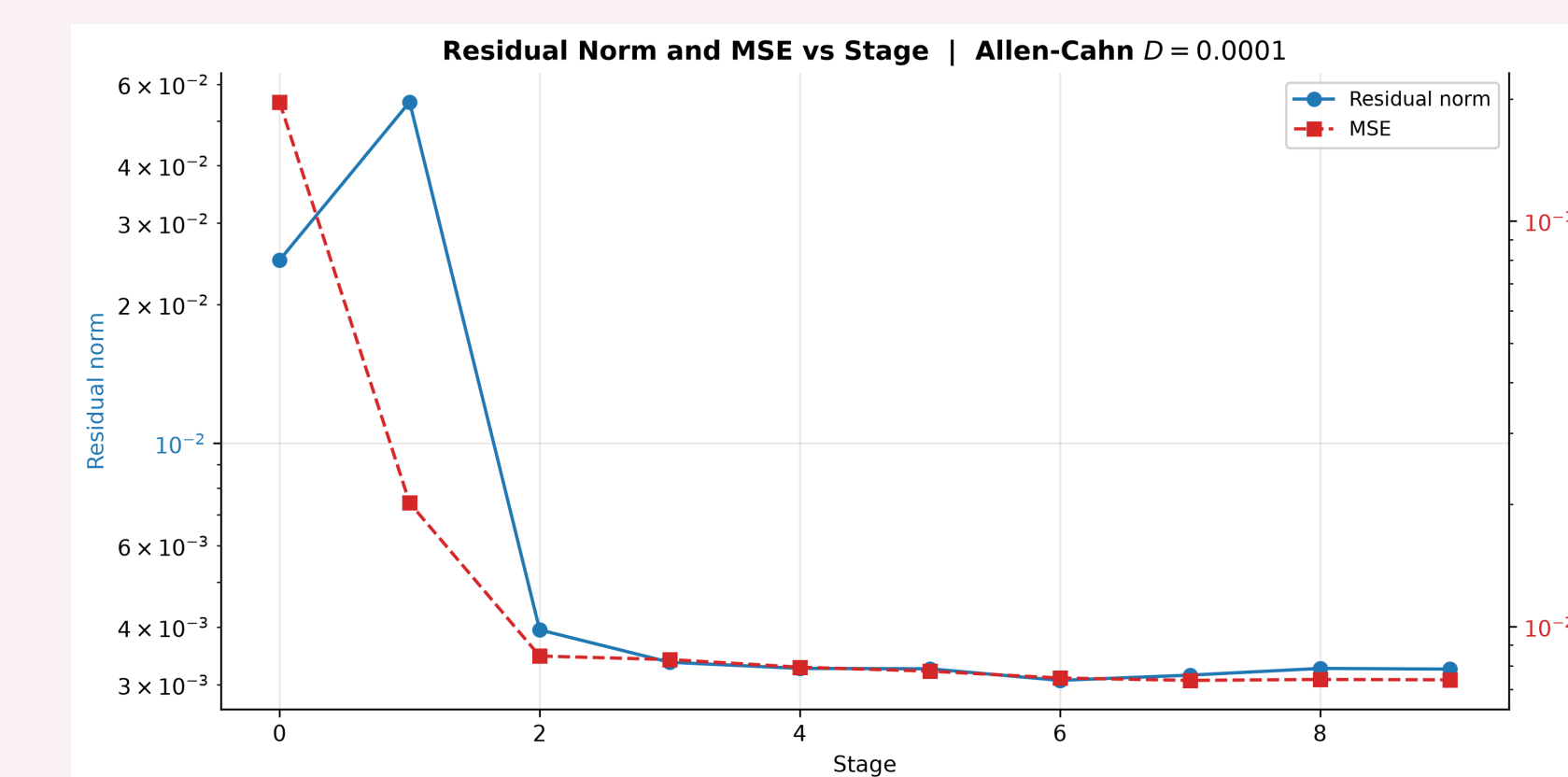


Figure 3. Residual and MSE across stages for Allen–Cahn PDE (D=0.0001).

Sensitivity & Limitations

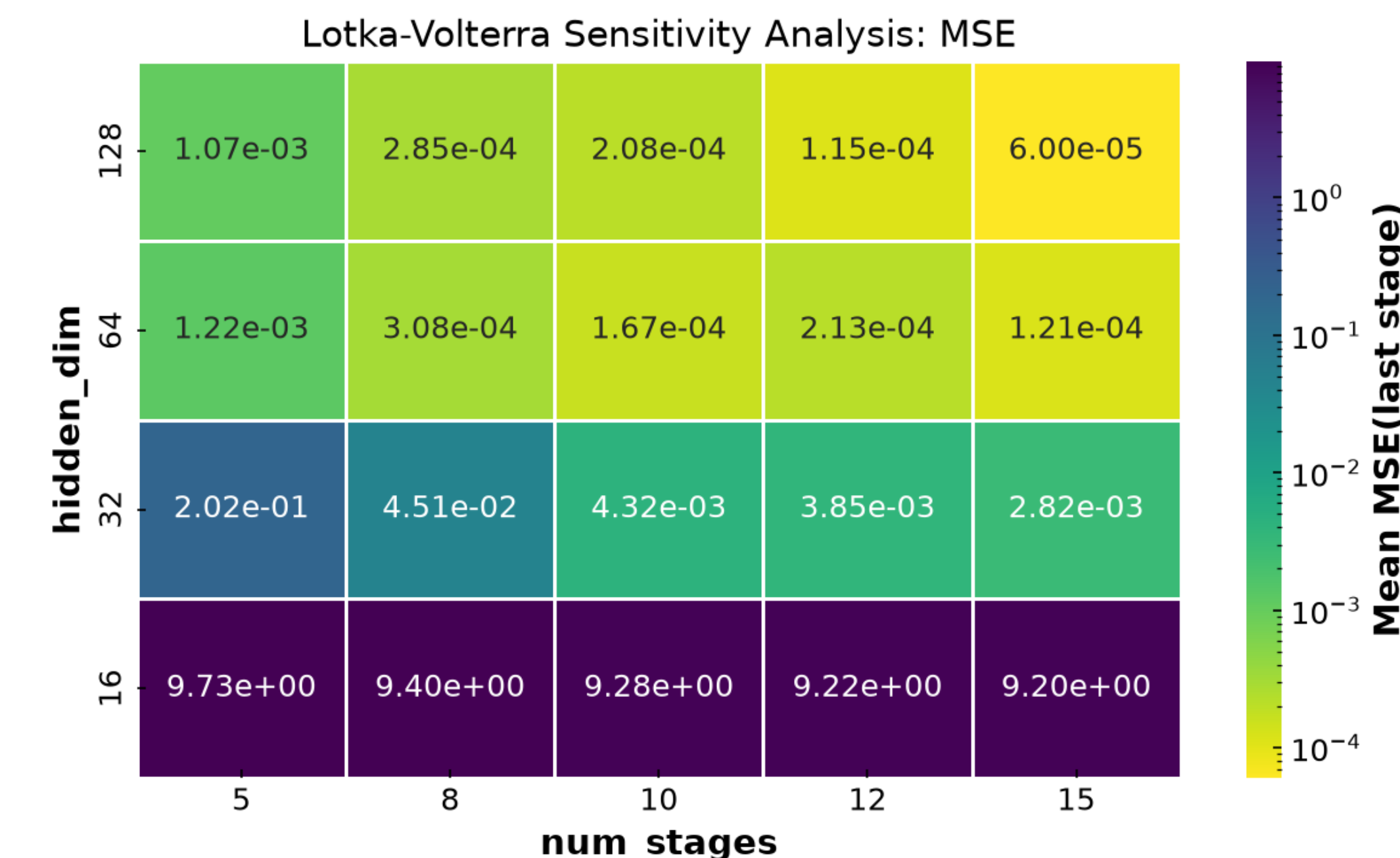


Figure 4. Sensitivity heatmap for Lotka–Volterra

- **More stages / larger weak learners** $\rightarrow$ lower MSE for nonstiff problems (e.g. Lotka–Volterra), but for stiff PDEs (Allen–Cahn) an **intermediate** hidden size (64–128) performs best — oversized weak learners re-introduce spectral bias.
- Boosting has a stiffness ceiling. Converges for Van der Pol when $\mu = 3$ but not for $\mu = 4$. Fails for Burgers' ($\nu = 0.002$). Boosted PINNs also suffer from long temporal issues as traditional PINNs.
- Second-order optimizers are currently validated only on **nonstiff ODEs**; extending to stiff PDEs is future work.

References

- [1] Rafael Bischof and Michael Kraus. Mixture-of-experts-ensemble meta-learning for physics-informed neural networks. 09 2022.
- [2] Zhiwei Fang, Sifan Wang, and Paris Perdikaris. Ensemble learning for Physics Informed Neural Networks: a Gradient Boosting approach. *arXiv (Cornell University)*, 2 2023.